

AK High-Dimensional Projection Structural Theory

Appendix CM

Core Corollary, Implementation Lemma, and Consumer-Contract Library

v21 synchronization of the frozen CM-v20 micro-theorem corpus
with derived closure, TB handoff repair, and dual-Core guard-rails

Theory version: v21.0.0

Appendix version: CM-v21.0.0

Canonical-ownership rule. AK-HPDST v21.0.0 Main Chapters 2–8 own the canonical generic Core statements. Foundation Appendices A–G own expanded Core-P proofs and record schemas; Appendix DG owns expanded Core-DG proofs. Appendix CM owns compact corollaries, implementation lemmas, counterexample witnesses, consumer contracts, and exact locators. Repetition never creates a competing Main theorem identity.

Immutable predecessor rule. The complete CM-v20 corpus, including its seventy-three local UIDs, remains an immutable lower-baseline source. Its three-mode comparison policy and four Core-P detector stages are historical v20 data. v21 consumption uses new v21 records and never retroactively rewrites those items.

Nonpromotion rule. A compact micro-theorem does not construct a finite diagram, prove descent effectivity, establish a Bridge or Return, repair a missing artifact, or promote a toy theorem. Readiness, verifier status, mathematical result, workflow completion, migration, and registration remain separate sorts.

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Abstract

Appendix CM-v21 is the compact corollary, implementation-lemma, and consumer-contract layer of AK-HPDST v21.0.0. It preserves the frozen seventy-three-item CM-v20 corpus at its exact historical strength while rebinding v21 use to the dual-Core constitution. Main Chapters 2–6 and Foundation Appendices A–G remain canonical for persistence metrics, detector certification, higher edges, tower defects, Restart, atlases, and finite-to-infinite reduction; Main Chapter 7 and Appendix DG remain canonical for finite diagrams and effective descent; Chapter 8 owns verifier and proof-DAG semantics; Chapter 9 owns Bridge and Return composition.

The v21 layer separates the four comparison modes, six detector stages, atomic verifier statuses, the four-valued P2DG readiness axis, later descent/Bridge handoff statuses, and target-indexed Bridge results. It supplies a lossless CM component contract, theorem-local metadata and register closure, a complete grouped migration map, a TB consumer handoff, and an explicit quarantine of nine dangling CM-v20 references found in the frozen TB-v20 artifact. Twelve Appendix-local derived results are added: threshold-band and window-meet actions, detector monotonicity and family-union bounds, exact augmented-mesh cardinality, associative Restart summaries, terminal-clean two-out-of-three and short-five principles, finite-chain defect accumulation, stabilization-witness independence, and two missing counterexample boundaries required by the TB lineage. These are CM-local consequences and do not displace their Main or Foundation premises.

No external-domain theorem, Core-DG reconstruction, object descent, HT/UB closure, final Claim UID, or release acceptance is asserted here.

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Part I. v21 constitutional synchronization and CM consumer architecture

CM.0.1 Scope, ownership, and conservative migration

v21 Definition 1 (CM-v21 role and authority). Appendix CM-v21 has the primary layer role

Core corollary and implementation library.

Its admissible local subtypes are compact corollary, implementation lemma, regression theorem, counterexample, consumer contract, serialization rule, and migration record. The canonical owners are:

- (i) Main Chapter 2 and Appendix A for metric, crop, threshold, comparison, and repaired-zero results;
- (ii) Main Chapter 3 and Appendix B for detector soundness, completeness, and certificate complexity;
- (iii) Main Chapter 4 and Appendix C for representatives, higher edges, and finite one-parameter reconstruction;
- (iv) Main Chapter 5 and Appendix D for tower morphisms, terminal/transient defects, derived limits, and finite-to-infinite reduction;
- (v) Main Chapter 6 and Appendix E for atlas, overlap, coherence, and Restart;
- (vi) Main Chapter 7 and Appendix DG for finite diagrams, cones, generators, Morita reconstruction, Alexandrov realization, and effective descent;
- (vii) Main Chapter 8 and Appendices F–G for atomic verification, proof-DAG closure, manifests, and governance;
- (viii) Main Chapter 9 for generic Bridge and Return composition.

A CM statement may be cited as a compact locator or implementation consequence but never as a second canonical owner of the governing Main theorem.

v21 Policy 2 (CM-v21-OWN-01: immutable predecessor and new-identity policy). Every CM-v20-NNN item remains available only through its frozen v20 source, hypotheses, proof, role, status, comparison policy, and prohibited stronger readings. A v21 restatement, strengthening, repair, or differently typed consumer contract receives a new v21 identity. Editorial merger or relocation is not theorem strengthening, and this appendix does not retroactively mutate the v20 Claim Register.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-OWN-01

Role / evidence: operational_policy / operational

Invocation / conformance: operational_rule / conformant

Premise authority / dependencies: Main Ch. 1 conservative-evolution and migration calculus

Comparison policy / detector-stage use: not_applicable/not_invoked

Output ceiling / failure routes: immutable-predecessor and new-identity discipline; FR-MISSING, FR-NONVERDICT, FR-T5

Migration relation: inherited; merged constitutional restatement with a new local identity.

v21 Definition 3 (Grouped migration disposition). The seventy-three predecessor items are migrated in the following owner groups:

Predecessor range	Canonical v21 owner	CM-v21 disposition
001	Main Chs. 1,8 / App. F–G	inherited; merged governance locator; new consumer schema is separately identified
002–018, 042–046	Main Ch. 2 / App. A	inherited; relocated, merged; compact metric/crop corollaries only
041, 047–056	Main Ch. 3 / App. B	inherited; relocated, merged; detector implementation locators
019–021, 032, 057–062	Main Ch. 4 / App. C	inherited; relocated, merged; higher-edge and representative locators
022–025, 031, 033–040, 063–067, 070–071	Main Ch. 5 / App. D	inherited; relocated, merged; tower/defect/reduction locators
026–030, 068–069, 072	Main Chs. 6,8 / Apps. E–F	inherited; relocated, merged; Restart/atlas/noncompensation locators
073	Main Ch. 8 / App. G	inherited; relocated, merged governance locator; v21 closure has a new identity

The external migration matrix records every item individually. No range assignment changes a predecessor theorem’s historical strength.

CM.0.2 Typed v21 consumption calculus

v21 Definition 4 (Seven-axis CM statement record). Every v21 CM item keeps independent fields for document role, evidence status, invocation role, conformance status, verifier status, workflow status, and migration status. A favorable value on one axis never promotes another. In particular, compiled, serialized, reviewed, migrated, or registered is not synonymous with proved or pass.

v21 Definition 5 (Four comparison modes). Every comparison consumed by a v21 CM component declares exactly one mode:

`exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared.`

The frozen CM-v20 trichotomy remains valid for v20 runs. A v21 use of `theorem_controlled_nonmetric` requires a new record naming the transport theorem, exact source and target types, direction, hypotheses, naturality scope, artifacts, and prohibited stronger readings. It is not inferred by editing the v20 mode token.

v21 Definition 6 (Six detector stages and cross-Core firewall). The v21 stage vocabulary is

`windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect, diagram_probe, descent_coherence.`

The first four are Core-P stages. The last two belong to Core-DG. A birth, germ, stencil, or family-mesh theorem proved at a Core-P stage is not transferred to `diagram_probe` or `descent_coherence` by renaming. Such use requires the Main Chapter 4/7 or Appendix DG reconstruction and conservativity theorem that supplies the actual diagrammatic object consumed.

v21 Definition 7 (Status-sort separation). The atomic verifier status is exactly one of

`pass, reject, undefined, not_invoked.`

A Chapter 2 P2DG input uses its own readiness axis

`ready, incomplete, inconsistent, not_invoked;`

there is no canonical token `p2dg_input_ready`. The later handoff statuses

`descent_input_ready, bridge_input_ready`

belong respectively to the Chapter 6–7 descent handoff and the Chapter 8–9 pre-Bridge handoff. None is an atomic verifier status.

For one fixed target record, the Chapter 9 Bridge-result sort is exactly

`reject`, `undefined`, `obstructed`, `partial_return`, `regularized_return`, `return`.

The lexical overlap of `reject` and `undefined` with atomic-status words does not identify the sorts: every serialized value carries its field and target. No token is copied across these axes.

v21 Policy 8 (CM-v21-IF-01: P2DG and Bridge nonpromotion). A CM detector, barcode, representative, Ext edge, Type IV record, Restart record, or finite reduction may prepare a P2DG or BridgeDAGInput input only when the governing Main/Foundation Interface has produced the complete required record. No readiness value or handoff status constructs a finite diagram, actual natural transformation, cone, cocycle, descended object, external target, or Return theorem.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-IF-01

Role / evidence: operational_policy / operational

Invocation / conformance: operational_rule / conformant

Premise authority / dependencies: Main Chs. 2,4,7–9; Appendices A–G and DG

Comparison policy / detector-stage use: runtime_resolved/not_invoked

Output ceiling / failure routes: P2DG and Bridge nonpromotion boundary; FR-MISSING, FR-NONVERDICT, FR-T3, FR-T5

Migration relation: inherited; merged cross-layer nonpromotion guard-rails.

CM.0.3 Lossless CM component and TB consumer contract

v21 Definition 9 (CM implementation component). A reusable CM implementation component is the record

$$\mathfrak{C}_{\text{CM}} = (\text{ID}, \text{owner}, C_{\text{src}}, C_{\text{tgt}}, \text{var}, \text{coeff/endpt}, \text{Hyp}, \text{mode}, \text{stage}, \text{Out}, \Gamma, \mathfrak{S}, \text{Fail}, \mathcal{A}, \text{Reg}).$$

It contains the immutable component identity, canonical theorem owner, source and target types, variance, coefficient and endpoint conventions, complete hypotheses, comparison mode, detector stage when invoked, exact output, propagated assumption context Γ , downward-closed supported conclusion set $\mathfrak{S}$, primary failure leaves, artifact bundle, and regression witnesses. Each invocation serializes its atomic status and any applicable readiness or target-result value in separate typed fields. Omitting a required field makes theorem-facing consumption undefined; a favorable sibling cannot fill the omission.

v21 Policy 10 (CM-v21-TB-01: TB consumer handoff). A TB-v21 theorem may consume a CM component only after checking exact component identity, canonical owner, source/target type, variance, coefficient and endpoint conventions, all hypotheses, comparison mode, detector stage, conclusion strength, primary failure route, and artifact identity. The TB theorem may specialize or instantiate the component but may not strengthen it, convert a toy realization into a general Interface, or cite document adjacency as a proof edge.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-TB-01

Role / evidence: consumer_contract / operational

Invocation / conformance: operational_rule / conformant

Premise authority / dependencies: CM implementation schema; frozen TB-v20 dependency discipline

Comparison policy / detector-stage use: runtime_resolved/not_invoked

Output ceiling / failure routes: typed CM-to-TB consumption contract; FR-MISSING, FR-NONVERDICT, FR-T5

Migration relation: strengthened successor to the implicit v20 consumer discipline; no predecessor UID is over-written.

v21 Policy 11 (CM-v21-TB-02: dangling v20 source-reference quarantine). The frozen TB-v20 artifact cites the nonexistent predecessor tokens

CM-v20-075, CM-v20-077, CM-v20-079,

CM-v20-080, CM-v20-081, CM-v20-082,

CM-v20-083, CM-v20-085, CM-v20-086.

The canonical CM-v20 source and final v20 register end at CM-v20-073. Therefore those citations are unresolved historical source bindings and are never fabricated as v20 UIDs. Future TB-v21 repair uses the explicit compatibility map below; the frozen TB-v20 text remains immutable.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-TB-02

Role / evidence: source_binding_repair_policy / operational

Invocation / conformance: operational_rule / conformant

Premise authority / dependencies: Frozen CM-v20 registry 001–073; frozen TB-v20 source bindings

Comparison policy / detector-stage use: not_applicable/not_invoked

Output ceiling / failure routes: quarantine of nine nonexistent predecessor tokens; FR-MISSING, FR-NONVERDICT, FR-T5

Migration relation: repaired binding route; the invalid predecessor references remain historical.

Dangling token	v21 repair source	Permitted repaired reading
075	CM-v20-051, 052 + CM-v21-DER-05	uniform-excess completeness plus a canonical minimal effective-domain stencil
077	CM-v20-057, 059, 060	synthetic shifted duality with separately declared natural zero reflection and nonpromotion
079	CM-v20-063, 067	terminal-zero/transient separation and counterexample boundary
080	CM-v21-DER-11	object equality or distance does not determine morphism defects
081	CM-v20-064	stage-typed complete transient-defect detection
082	CM-v21-DER-12	local-zero/global-nonzero window-partition counterexample
083	CM-v20-068	retained-bar coverage required for atlas globalization
085	CM-v20-059, 062	natural zero-reflection and complete-detector higher-edge discharge
086	CM-v20-073 + CM-v21-GOV-01	manifest/source-binding closure; registration remains nonproof

v21 Remark 12 (CM-v21-SRC-01: imported single-morphism source-binding audit). The imported statement used by CM-v20-038 was rechecked against Bauer–Lesnick, *Induced Matchings and the Algebraic Stability of Persistence Barcodes*, Journal of Computational Geometry 6(2) (2015), Corollary 6.6, DOI 10.20382/jocg.v6i2a9. The exact licensed statement is: for one-parameter pointwise finite-dimensional persistence modules M, N , they are δ -interleaved if and only if there exists $f : M \rightarrow N(\delta)$ whose kernel and cokernel are both 2δ -trivial. CM does not promote this locator to an unchanged multidimensional converse, an arbitrary q-tame single-morphism theorem, or a statement outside the hypotheses printed by the source.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-SRC-01

Role / evidence: source_binding_audit / operational

Invocation / conformance: provenance_only / conformant

Premise authority / dependencies: Bauer–Lesnick 2015, Corollary 6.6

Comparison policy / detector-stage use: not_compared/not_invoked

Output ceiling / failure routes: exact one-parameter p.f.d. source-strength locator; FR-MISSING, FR-NONVERDICT, FR-T5

Migration relation: inherited; source-only verification of the imported locator.

Part II. CM-v21 Appendix-local derived micro-theorems

v21 Theorem 13 (CM-v21-DER-01: threshold-deletion band action). For positive thresholds, the operators $\{T_\tau^{\text{bar}}\}_{\tau>0}$ form a commutative idempotent semigroup under composition:

$$T_\sigma^{\text{bar}} T_\tau^{\text{bar}} = T_\tau^{\text{bar}} T_\sigma^{\text{bar}} = T_{\max\{\tau, \sigma\}}^{\text{bar}}, \quad (T_\tau^{\text{bar}})^2 = T_\tau^{\text{bar}}.$$

Moreover, if $\tau \leq \sigma$, then $T_\sigma^{\text{bar}}(B)$ is the retained subprofile of $T_\tau^{\text{bar}}(B)$ obtained by deleting the remaining finite bars of length at most σ . This is an operator-algebra statement at barcode-profile level, not exactness or functoriality on a filtered or derived category.

Proof. The composition formula is CM-v20-004 and is symmetric in τ, σ ; idempotence is CM-v20-003. The final statement is CM-v20-005 applied barwise. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-01

Role / evidence: implementation_lemma / proved

Invocation / conformance: theorem_evidence / conformant

Premise authority / dependencies: CM-v20-003, CM-v20-004, CM-v20-005

Comparison policy / detector-stage use: not_compared/not_invoked

Output ceiling / failure routes: exact barcode-profile semilattice action; no categorical exactness; FR-MISSING, FR-T5

Migration relation: strengthened; merged exact logical closure of CM-v20-003–005.

v21 Theorem 14 (CM-v21-DER-02: window-restriction meet action). For right-open windows U, V , including the empty intersection convention,

$$W_U W_V(B) = W_{U \cap V}(B) = W_V W_U(B)$$

as barcode multisets with multiplicity. Hence window restriction is an idempotent commutative meet action. No commutation with hard threshold deletion is inferred.

Proof. For each interval summand I , one has $(I \cap V) \cap U = I \cap (U \cap V)$. Empty clipped intervals are discarded on both sides, and multiplicities are unchanged. The warning follows from CM-v20-009. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-02

Role / evidence: implementation_lemma / proved

Invocation / conformance: theorem_evidence / conformant

Premise authority / dependencies: CM-v20-007, CM-v20-009

Comparison policy / detector-stage use: not_compared/not_invoked

Output ceiling / failure routes: exact window-intersection identity; no threshold commutation; FR-MISSING, FR-T3, FR-T5

Migration relation: strengthened; merged exact window calculus from CM-v20-007.

v21 Theorem 15 (CM-v21-DER-03: upward closure of complete stencils). Fix one detector stage, window, threshold, endpoint convention, cofinal policy, and declared profile family. If an admissible finite stencil T is coverage-complete and $T \subseteq T'$, where T' is another admissible finite stencil in the same effective domain, then T' is coverage-complete. Thus the set of complete finite stencils is upward closed under inclusion, while its inclusion-minimal members form an antichain.

Proof. Every retained bar has an active-start point in T ; the same point belongs to T' . CM-v20-050 gives zero reflection. Inclusion-minimal members of any upward-closed family are pairwise incomparable. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-03

Role / evidence: detector_corollary / proved

Invocation / conformance: theorem_evidence / conformant

Premise authority / dependencies: CM-v20-050

Comparison policy / detector-stage use: not_compared/windowed_prethreshold

Output ceiling / failure routes: coverage-completeness monotonicity under stencil inclusion; FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Migration relation: strengthened; merged detector consequence.

v21 *Theorem 16* (CM-v21-DER-04: family-union query bound in one fixed stencil model). Fix one exact non-adaptive point-stencil information/query model, one detector stage, window, threshold, endpoint convention, cofinal policy, encoding convention, and verifier semantics. Let

$$q_{\min}^{\text{stc}}(\mathcal{F})$$

denote the legacy worst-case query-coordinate minimum for the repaired-zero decision problem on a declared family $\mathcal{F}$ in this fixed model. If finite stencils T_i are complete for $\mathcal{F}_i$, then $T_1 \cup T_2$ is complete for $\mathcal{F}_1 \cup \mathcal{F}_2$, and whenever the two minima are finite,

$$q_{\min}^{\text{stc}}(\mathcal{F}_1 \cup \mathcal{F}_2) \leq q_{\min}^{\text{stc}}(\mathcal{F}_1) + q_{\min}^{\text{stc}}(\mathcal{F}_2).$$

No corresponding scalar bound for bit length, verifier time, artifact footprint, or the full v21 Pareto frontier is inferred without a model-specific composition theorem.

Proof. A retained bar in a member of the union belongs to at least one family and is detected by the corresponding stencil. The union therefore gives a valid complete scheme, and its number of distinct queries is at most the sum of the two query counts. Taking the minimum over all complete stencils proves the inequality. The final boundary follows because the other certificate-cost coordinates need not be additive and may involve encoding, verification, or artifact-reuse policies absent from the hypothesis. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-04

Role / evidence: certificate_complexity_corollary/proved

Invocation / conformance: theorem_evidence/conformant

Premise authority / dependencies: CM-v20-050, CM-v20-053; Main Ch. 3 fixed-model cost semantics

Comparison policy / detector-stage use: not_compared/windowed_prethreshold

Output ceiling / failure routes: legacy exact-nonadaptive query-coordinate upper bound only; FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Migration relation: strengthened fixed-model query-coordinate consequence.

v21 *Theorem 17* (CM-v21-DER-05: canonical minimal augmented-mesh stencil and refinement). Let $W = [a, b]$ be bounded with $b - a > \tau$, put

$$c = b - \tau, \quad W_\tau = [a, c), \quad L_{\text{eff}} = c - a > 0,$$

and fix $h > 0$. Set $N = \lceil L_{\text{eff}}/h \rceil$ and define the interior stencil

$$T_h = \left\{ a + \frac{jL_{\text{eff}}}{N} : 1 \leq j \leq N - 1 \right\} \subseteq [a, c).$$

Then

$$|T_h| = N - 1 = \max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{h} \right\rceil - 1 \right\}, \quad \text{Mesh}_{W_\tau}(T_h) = \frac{L_{\text{eff}}}{N} \leq h.$$

This cardinality is minimal among finite stencils with augmented mesh at most h . Consequently, if a declared family has an independently certified retained-bar excess $e_{\tau, W}(\mathcal{F}) > h$, then T_h is complete for that family, and every refinement remains complete. If $b - a \leq \tau$, the effective domain is empty and the finite-bar zero conclusion uses the separate trivial lane rather than boundary points treated as detector samples.

Proof. The $N - 1$ points divide $[a, c]$ into N equal augmented gaps, each of length $L_{\text{eff}}/N \leq h$. Conversely, m interior points create $m + 1$ augmented gaps summing to L_{eff} ; if every gap is at most h , then $m + 1 \geq L_{\text{eff}}/h$, hence $m \geq \lceil L_{\text{eff}}/h \rceil - 1$. Family completeness follows from CM-v20-051, and refinement from CM-v21-DER-03. The last sentence is the empty-effective-domain case of the canonical family-mesh theorem. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-05
Role / evidence: detector_implementation_theorem / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-051, CM-v20-052; Main Ch. 3 augmented-mesh convention
Comparison policy / detector-stage use: not_compared/windowed_prethreshold
Output ceiling / failure routes: minimal interior stencil for a prescribed augmented-mesh ceiling; FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Migration relation: strengthened; merged and relocated explicit construction.

v21 Theorem 18 (CM-v21-DER-06: associative Restart summary calculus). A certified finite Restart block with inequalities $r_{j+1} \geq \kappa_j r_j - \eta_j$ has the affine summary

$$(K, E) = \left(\prod_j \kappa_j, \sum_m \left(\prod_{j>m} \kappa_j \right) \eta_m \right), \quad r_{\text{out}} \geq K r_{\text{in}} - E.$$

If consecutive blocks have summaries (K_1, E_1) and (K_2, E_2) , their composite summary is

$$(K_2, E_2) \star (K_1, E_1) = (K_2 K_1, K_2 E_1 + E_2).$$

The operation $\star$ is associative with identity $(1, 0)$. Thus any finite Restart chain may be compressed into blocks without changing the proved lower bound, provided all non-scalar handoff obligations remain attached.

Proof. Substitution gives $r_2 \geq K_2(K_1 r_0 - E_1) - E_2$. A direct calculation proves associativity and the identity law. Expanding the iterated product recovers CM-v20-026. The final qualification is CM-v20-028 and CM-v20-069. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-06
Role / evidence: Restart_implementation_lemma / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-026, CM-v20-028, CM-v20-069
Comparison policy / detector-stage use: not_compared/not_invoked
Output ceiling / failure routes: associative affine summary; non-scalar obligations retained; FR-MISSING, FR-T3, FR-T5
Migration relation: strengthened; merged finite-chain Restart consequence.

v21 Theorem 19 (CM-v21-DER-07: terminal-clean two-out-of-three). Let $M \xrightarrow{f} N \xrightarrow{g} P$ be actual persistence morphisms on a common terminal scope. Assume the source and target modules and the defect profiles of f, g, gf are terminal-tame. Write $\text{Clean}_{\text{tg}}(h)$ for $\mu(h) = \nu(h) = 0$. Any two of

$$\text{Clean}_{\text{tg}}(f), \quad \text{Clean}_{\text{tg}}(g), \quad \text{Clean}_{\text{tg}}(gf)$$

imply the third.

Proof. After one common terminal parameter the three morphisms become finite-dimensional linear maps. Terminal cleanliness is equivalent to the stabilized map being an isomorphism. Isomorphisms satisfy two-out-of-three. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-07
Role / evidence: Type_IV_corollary / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-023, CM-v20-036
Comparison policy / detector-stage use: not_compared/not_invoked
Output ceiling / failure routes: terminal-clean two-out-of-three on one common terminal scope; FR-MISSING, FR-T4, FR-T5
Migration relation: strengthened; merged terminal-linear consequence.

v21 Theorem 20 (CM-v21-DER-08: terminal short-five principle). Consider a commutative diagram with short exact rows and vertical maps ϕ', ϕ, ϕ'' . Suppose all six modules and all three vertical defect profiles are terminal-tame on one common terminal scope. If any two vertical maps are terminal-clean, then the third is terminal-clean.

Proof. Evaluate after a common stabilization parameter. The rows become short exact sequences of finite-dimensional vector spaces. If the two outer maps are isomorphisms, the short five lemma gives the middle isomorphism. If the middle and right maps are isomorphisms, identify the left terms as kernels; if the left and middle maps are isomorphisms, identify the right terms as cokernels. In each case the remaining stabilized map is an isomorphism, hence has zero terminal kernel and cokernel. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-08
Role / evidence: Type_IV_corollary / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-037; exactness on a common terminal scope
Comparison policy / detector-stage use: not_compared/not_invoked
Output ceiling / failure routes: short-exact terminal-clean two-out-of-three; FR-MISSING, FR-T4, FR-T5
Migration relation: strengthened; merged short-exact consequence.

v21 Theorem 21 (CM-v21-DER-09: finite-chain accumulation of Type IV defects). Let

$$M_0 \xrightarrow{f_1} M_1 \xrightarrow{f_2} \dots \xrightarrow{f_m} M_m$$

be actual morphisms on a common terminal scope, and assume every f_i and every partial composite has terminal-tame defects. Then

$$\mu(f_m \cdots f_1) \leq \sum_{i=1}^m \mu(f_i), \quad \nu(f_m \cdots f_1) \leq \sum_{i=1}^m \nu(f_i).$$

In particular, a finite composite of terminal-clean morphisms is terminal-clean.

Proof. Induct on m using CM-v20-036 at each composition. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-09
Role / evidence: Type_IV_implementation_lemma / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-036
Comparison policy / detector-stage use: not_compared/not_invoked
Output ceiling / failure routes: finite-chain subadditive terminal-defect bound; FR-MISSING, FR-T4, FR-T5
Migration relation: strengthened; merged finite-chain consequence.

v21 Theorem 22 (CM-v21-DER-10: stabilization-witness independence). Let a directed system be eventually constant, and let $N \leq N'$ be two certified stabilization stages. Suppose the stable transition $M_N \rightarrow M_{N'}$ and the two apex agreements are compatible with the same declared tail cocone. Then the colimit-to-apex isomorphisms constructed from N and N' coincide under the canonical stable-stage identifications. Consequently the exact finite-to-infinite verdict is independent of the chosen certified stabilization witness.

Proof. Both maps are the unique map from the colimit induced by the same apex cocone. The universal property forces equality. CM-v20-070 supplies the isomorphism conclusion. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-10
Role / evidence: finite_to_infinite_corollary / proved
Invocation / conformance: theorem_evidence / conformant
Premise authority / dependencies: CM-v20-070
Comparison policy / detector-stage use: exact/not_invoked
Output ceiling / failure routes: choice independence under one compatible apex cocone; FR-MISSING, FR-T4, FR-T5

Migration relation: strengthened; merged universal-property consequence.

v21 Theorem 23 (CM-v21-DER-11: object comparison does not determine morphism defects). Object-level equality, isomorphism, or zero bottleneck distance of sources and targets does not determine the kernel or cokernel of a comparison morphism. For every nonzero finite persistence module M , the maps

$$\text{id}_M : M \rightarrow M, \quad 0_M : M \rightarrow M$$

have identical endpoint objects, while the first has zero defects and the second has kernel and cokernel isomorphic to M .

Proof. This is immediate from the definitions of kernel and cokernel. Defect information belongs to the actual morphism, not merely its endpoints. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-11

Role / evidence: regression_counterexample / proved

Invocation / conformance: theorem_evidence / conformant

Premise authority / dependencies: TB-v20-047 mathematical body; abelian profile-category kernel/cokernel calculus

Comparison policy / detector-stage use: not_compared/not_invoked

Output ceiling / failure routes: no endpoint-object-only defect inference; FR-MISSING, FR-T5

Migration relation: inherited; relocated from frozen TB-v20-047.

v21 Theorem 24 (CM-v21-DER-12: local-zero/global-nonzero window witness). Let $M = k_{[0,2)}$, $\tau = 3/2$, $W_1 = [0, 1)$, and $W_2 = [1, 2)$. Then

$$T_\tau^{\text{bar}} W_{W_1}(M) = 0, \quad T_\tau^{\text{bar}} W_{W_2}(M) = 0, \quad T_\tau^{\text{bar}}(M) = M \neq 0.$$

Thus simultaneous local repaired zero on a covering partition does not imply global repaired zero without retained-bar coverage or a valid detector-aware Restart/globalization theorem.

Proof. Each clipped interval has length $1 \leq 3/2$ and is deleted, while the global interval has length $2 > 3/2$ and is retained. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-DER-12

Role / evidence: regression_counterexample / proved

Invocation / conformance: theorem_evidence / conformant

Premise authority / dependencies: CM-v20-009, CM-v20-068, CM-v20-069; TB-v20-049 body

Comparison policy / detector-stage use: not_compared/not_invoked

Output ceiling / failure routes: local repaired zero does not globalize without coverage; FR-MISSING, FR-DETECTO R-CERTIFICATION, FR-T3, FR-T5

Migration relation: inherited; relocated from frozen TB-v20-049.

v21 Policy 25 (CM-v21-GOV-01: lossless serialization and package closure). The v21 CM package serializes: the frozen source identity; all seventy-three predecessor UIDs; every v21 local identity and theorem-local metadata block; canonical owner; migration status and modifiers; source/target types; variance; coefficient and endpoint conventions; complete hypotheses; comparison mode; detector stage; atomic status; the P2DG readiness value or later handoff status when applicable; the target-indexed Bridge result when applicable; propagated assumption context; supported strength set; primary failure routes; artifact identities; TB compatibility mapping; and explicit Claim-Register deferral. Serialization, compilation, replay, or hash agreement verifies identity and auditability only; it cannot establish a missing mathematical premise.

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-GOV-01

Role / evidence: serialization_policy / operational

Invocation / conformance: operational_rule / conformant

Premise authority / dependencies: CM-v20-073; Main Ch. 8 and Appendix G

Comparison policy / detector-stage use: runtime_resolved/not_invoked

Output ceiling / failure routes: lossless package and status-sort serialization; FR-MISSING, FR-NONVERDICT, FR-T5

Migration relation: strengthened successor to CM-v20-073; merged v21 serialization fields.

v21 Theorem 26 (CM-v21-CLS-01: conservative CM-v21 closure). Assume the canonical Main/Foundation premises cited by each consumed component, the complete CM implementation record, dependency-closed atomic pass statuses, and lossless artifact serialization. Then Appendix CM-v21 supplies a reusable compact locator or implementation consequence at exactly the displayed output strength. It supplies no automatic CoreP-to-CoreDG reconstruction, descent effectivity, HT/UB composition, external Return, final Claim UID, or release acceptance.

Proof. Each CM conclusion follows from its displayed Main/Foundation premise or Appendix-local proof, while the component schema prevents omitted hypotheses or strength promotion. The excluded conclusions belong to separately typed Interface, Bridge, Return, or governance owners and therefore do not follow from CM closure. $\square$

Local UID: ak-hpdst:21.0.0:CM-v21:CM-v21-CLS-01

Role / evidence: package_closure_theorem / conditional

Invocation / conformance: theorem_evidence / conformant_with_hypotheses

Premise authority / dependencies: all invoked Main/Foundation premises and CM component records

Comparison policy / detector-stage use: runtime_resolved/not_invoked

Output ceiling / failure routes: compact locator/implementation consequence at supported strength only; FR-MISSING, FR-NONVERDICT, FR-T3, FR-T4, FR-T5

Migration relation: strengthened; merged v21 closure consequence.

CM.0.4 Complete v21 local register summary

The following register lists every stable v21 local identity in this appendix. Full hypotheses, output ceilings, failure routes, and migration relations remain controlled by the adjacent metadata blocks and immutable source.

Local ID	Role	Evidence	Premise authority / dependencies
CM-v21-OWN-01	operational policy	operational	Main Ch. 1 migration calculus
CM-v21-IF-01	operational policy	operational	Main Chs. 2, 4, 7–9; A–G, DG
CM-v21-TB-01	consumer contract	operational	CM schema; frozen TB dependency discipline
CM-v21-TB-02	binding repair policy	operational	CM-v20 001–073; frozen TB bindings
CM-v21-SRC-01	source-binding audit	operational	Bauer–Lesnick 2015, Cor. 6.6
CM-v21-DER-01	implementation lemma	proved	CM-v20-003–005
CM-v21-DER-02	implementation lemma	proved	CM-v20-007, 009
CM-v21-DER-03	detector corollary	proved	CM-v20-050
CM-v21-DER-04	fixed-model query bound	proved	CM-v20-050, 053; Main Ch. 3 model semantics
CM-v21-DER-05	detector construction	proved	CM-v20-051, 052; Main augmented mesh
CM-v21-DER-06	Restart summary lemma	proved	CM-v20-026, 028, 069
CM-v21-DER-07	Type IV corollary	proved	CM-v20-023, 036
CM-v21-DER-08	short-exact corollary	proved	CM-v20-037; common terminal scope
CM-v21-DER-09	Type IV chain bound	proved	CM-v20-036
CM-v21-DER-10	reduction corollary	proved	CM-v20-070
CM-v21-DER-11	regression counterexample	proved	TB-v20-047 body; defect calculus
CM-v21-DER-12	regression counterexample	proved	CM-v20-009, 068, 069; TB-v20-049
CM-v21-GOV-01	serialization policy	operational	CM-v20-073; Main Ch. 8; Appendix G
CM-v21-CLS-01	closure theorem	conditional	invoked Main/Foundation and CM records

v21 Remark 27 (Register-count and Claim-Register boundary). This source contains nineteen stable v21 local identities: five operational or source-binding controls, twelve derived mathematical statements, one serialization policy, and one conditional closure theorem. This local register is not the final v21 Claim Register and creates no Claim UID or release acceptance.

Part III. Frozen CM-v20 predecessor corpus

The following numbered sections and CM-v20-001–CM-v20-073 items are retained as the immutable predecessor source. Their historical titles, roles, hypotheses, proofs, comparison policies, and UUIDs are not rewritten into v21 claims. The only textual repair is the correction of a literal transcription tab in the printed token τ in CM-v20-043 and its registry row; this does not change the mathematical statement.

1 Scope, dependency baseline, and claim discipline

Policy 1.1 (CM-v20-001: Core micro-theorem boundary and no external upgrade). A CM-v20 item may be consumed only with its CM UID, declared role, evidence status, comparison policy, complete hypotheses, source Claim UUIDs, artifact identities, and failure routes. A CM-v20 theorem is internal to the declared v20 Core regime and yields no external-domain conclusion without a separately proved and registered Interface or Bridge Theorem.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-001
Role / status: operational_policy / operational
Source claims: CR-G-001, CR-G-002, CR-G-017, CR-G-018
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-001; inherited at immutable v19 strength.

Definition 1.2 (Standing repaired Core regime). Fix a field k . Work in the constructible one-parameter regime $\text{Pers}_k^{\text{cons}}$. For a finite-type filtered object F , monitored degree i , right-open window W , and threshold $\tau > 0$, define

$$B_{i,W}(F) := W_W P_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

The standard bottleneck metric with the inherited repaired finite/infinite-bar convention is used whenever metric comparison is invoked. Every detector record additionally declares one source stage among

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect.

A theorem proved at one stage is not silently transported to another stage.

Definition 1.3 (Typed CM status). Every CM item records a document role, evidence status, invocation role, conformance status, claim-level comparison policy, admissible runtime modes, source binding, and structured failure routes. Missing evidence gives undefined; it is never silently encoded as a numbered obstruction, zero, pass, not_invoked, or not_compared.

2 Barcode algebra and categorical guard-rails

Lemma 2.1 (CM-v20-002: Infinite-bar threshold invariance). *For every finite $\tau > 0$,*

$$T_\tau^{\text{bar}}([a, \infty)) = [a, \infty).$$

An infinite bar never lies in a finite threshold band $[\tau - \varepsilon, \tau + \varepsilon]$.

Proof. Its threshold length is $+\infty$, while hard deletion removes only finite bars of length at most τ . □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-002
Role / status: core_lemma / proved
Source claims: CR-A-005, CR-A-010; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-002; inherited at immutable v19 strength.

Theorem 2.2 (CM-v20-003: Hard threshold deletion idempotence). *For every barcode profile B and $\tau > 0$,*

$$T_{\tau}^{\text{bar}} T_{\tau}^{\text{bar}}(B) = T_{\tau}^{\text{bar}}(B).$$

Proof. After the first deletion no finite bar of length at most τ remains. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-003
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-009, CR-A-010
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-003; inherited at immutable v19 strength.

Theorem 2.3 (CM-v20-004: Hard threshold composition). *For $\tau, \sigma > 0$,*

$$T_{\sigma}^{\text{bar}} T_{\tau}^{\text{bar}}(B) = T_{\max\{\tau, \sigma\}}^{\text{bar}}(B).$$

Proof. The composite deletes exactly the finite bars whose length is at most τ or at most σ , equivalently at most $\max\{\tau, \sigma\}$. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-004
Role / status: core_micro_theorem / proved
Source claims: CR-CH1-018, CR-A-010; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-004; inherited at immutable v19 strength.

Theorem 2.4 (CM-v20-005: Retained-profile threshold monotonicity). *If $0 < \tau \leq \sigma$, every bar retained at threshold σ is retained at threshold τ .*

Proof. A finite length greater than σ is greater than τ ; infinite bars survive both thresholds. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-005
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-009; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-005; inherited at immutable v19 strength.

Theorem 2.5 (CM-v20-006: Positive-threshold short bars are not Serre). *For $\tau > 0$, the class of constructible persistence modules whose bars all have finite length at most τ is not extension-closed in general. Hence hard positive-threshold deletion is not globally licensed as a Serre localization or exact reflector.*

Proof. At $\tau = 1$, the exact sequence

$$0 \rightarrow k_{[1,2)} \rightarrow k_{[0,2)} \rightarrow k_{[0,1)} \rightarrow 0$$

has short subobject and quotient but a length-two middle interval. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-006
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-006, CR-A-008, CR-A-009
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-006; inherited at immutable v19 strength.

3 Window restriction, crop order, and repaired evaluation

Theorem 3.1 (CM-v20-007: Nested window restriction identity). *If $W' \subseteq W$, then at barcode-profile level*

$$W_{W'}W_W(B) = W_{W'}(B).$$

In particular, W_W is idempotent.

Proof. For each bar I , $(I \cap W) \cap W' = I \cap W'$. Empty intersections and multiplicities agree. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-007
Role / status: core_micro_theorem / proved
Source claims: CR-CH1-022, CR-A-031; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-007; inherited at immutable v19 strength.

Policy 3.2 (CM-v20-008: Fixed window-first repaired evaluation). The canonical evaluation order is

$$F \mapsto P_i(F) \mapsto W_W P_i(F) \mapsto T_\tau^{\text{bar}}(W_W P_i(F)).$$

Threshold clearance is measured after windowing and before deletion. A crop-after-collapse expression is not substituted unless CM-v20-010 is discharged.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-008
Role / status: operational_policy / operational
Source claims: CR-CH1-023, CR-CH2-033, CR-A-037
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-008; inherited at immutable v19 strength.

Theorem 3.3 (CM-v20-009: No unconditional crop-collapse commutation). *In general,*

$$W_W T_\tau^{\text{bar}}(B) \neq T_\tau^{\text{bar}} W_W(B).$$

Proof. Take $B = \{[0, 10)\}$, $W = [0, 1)$, and $\tau = 2$. Deletion first retains the long bar and cropping produces $[0, 1)$; cropping first produces a length-one bar which is deleted. The same phenomenon occurs for an infinite bar on a bounded window. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-009
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-028, CR-A-032
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-009; inherited at immutable v19 strength.

Theorem 3.4 (CM-v20-010: Exact crop–collapse criterion). *Let (B, W) be τ -window-adapted: every globally retained bar, including every infinite bar, is either disjoint from W or remains retained after clipping. Then, and only then,*

$$W_W T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}} W_W(B)$$

as barcode multisets including multiplicity.

Proof. Globally deleted finite bars remain deleted after clipping because clipping does not increase length. A globally retained bar contributes equally in both orders precisely under the stated all-retained-bars condition. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-010
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH2-031, CR-CH2-032, CR-A-035, CR-A-036
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-010; inherited at immutable v19 strength.

4 Metric repair and comparison-mode calculus

Theorem 4.1 (CM-v20-011: Global non-expansiveness of hard deletion fails). *There exist barcodes B, C with*

$$d_B(B, C) < d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C).$$

Proof. Fix $0 < \eta < \tau/4$ and let

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Then $d_B(B, C) = 2\eta$. After deletion only the bar of B remains, so the distance is $(\tau + \eta)/2 > 2\eta$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-011
Role / status: metric_repair_theorem / proved
Source claims: CR-CH2-010, CR-A-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-011; inherited at immutable v19 strength.

Theorem 4.2 (CM-v20-012: Sharp two-sided threshold-gap-safe stability). *Let B, C be finite barcodes, possibly with infinite bars. If $d_B(B, C) \leq \varepsilon$ and no finite bar of either barcode has length in the closed band*

$$[\tau - \varepsilon, \tau + \varepsilon],$$

then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The half-width ε is sharp for this two-sided certificate.

Proof. Choose an ε -matching. Infinite bars match only infinite bars and survive. A retained finite bar has length greater than $\tau + \varepsilon$. If its finite match were deleted, the two-sided closed-band hypothesis would force the matched length below $\tau - \varepsilon$, contradicting the fact that matched finite-bar lengths differ by at most 2ε . The argument is symmetric. Removing pairs deleted on both sides restricts the original matching to the retained barcodes with no larger cost. Sharpness follows from centered bars of lengths $\tau + c + \eta$ and $\tau - c - \eta$ for any $0 < c < \varepsilon$ and sufficiently small $\eta > 0$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-012
Role / status: metric_repair_theorem / proved
Source claims: CR-CH2-016, CR-A-017
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-012; inherited at immutable v19 strength.

Theorem 4.3 (CM-v20-013: Scalarized-ledger-safe threshold classification). *Let $\mathbf{d}_{\text{met}}$ lie in the declared metric-ledger quantale, let $\varepsilon_B = \text{val}_B(\mathbf{d}_{\text{met}})$, and suppose the record proves*

$$d_B(B, C) \leq \varepsilon_B \quad \text{and} \quad \varepsilon_B < \text{gap}_\tau(B, C).$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B.$$

Proof. The strict inequality gives two-sided avoidance of the closed threshold band of half-width ε_B . Apply CM-v20-012. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-013
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH2-017, CR-A-018
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-013; inherited at immutable v19 strength.

Proposition 4.4 (CM-v20-014: Exact comparison is not a positive-radius robustness certificate). *An exact equality or profile isomorphism between two specified profiles does not imply a positive threshold-gap radius.*

Proof. A profile containing a finite bar of length exactly τ is exactly equal to itself, but has $\text{clear}_\tau = 0$. Thus exact comparison and positive-radius robustness are logically distinct. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-014
Role / status: core_micro_theorem / proved
Source claims: CR-CH1-025, CR-CH1-026, CR-CH3-009
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-014; inherited at immutable v19 strength.

Policy 4.5 (CM-v20-015: Runtime comparison-mode trichotomy). Every consumed comparison resolves exactly one runtime mode:

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared}.$$

The first requires an actual equality or profile isomorphism; the second requires a pre-threshold discrepancy bound and two-sided threshold-gap record; the third is admissible only when no comparison inference is consumed and cannot discharge an overlap, transport, coherence, representative, tower, or Restart obligation.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-015
Role / status: operational_policy / operational
Source claims: CR-CH1-025, CR-CH7-028, CR-CH8-010
Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-015; inherited at immutable v19 strength.

Theorem 4.6 (CM-v20-016: Overlap transfer under a scalarized threshold-gap certificate). *For directly windowed pre-threshold overlap profiles $B_{i,W_{\alpha\beta}}^\alpha$ and $B_{i,W_{\alpha\beta}}^\beta$, suppose*

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

Then the repaired overlap profiles have bottleneck, and hence interleaving, distance at most $\varepsilon_{\alpha\beta,i}$ in the declared constructible regime.

Proof. Apply CM-v20-013 to the two directly windowed pre-threshold profiles and then the one-parameter isometry theorem. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-016
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH3-022, CR-A-041, CR-E-048
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-016; inherited at immutable v19 strength.

5 Representative and Ext-edge micro-theorems

Theorem 5.1 (CM-v20-017: Windowed representative idempotence). *Let $C_{\tau,W}F$ be a verified window-normalized representative. If a second representative is constructed with the same window, monitored scope, and profile-equivalence policy, then*

$$C_{\tau,W}(C_{\tau,W}F) \simeq_{\text{PH}, I_{\text{mon}}} C_{\tau,W}F.$$

This is profile-level equivalence, not strict filtered equality or a global idempotent endofunctor.

Proof. Window normalization prevents further clipping, and CM-v20-003 makes threshold deletion idempotent on every monitored profile. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-017
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-B-064
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-017; inherited at immutable v19 strength.

Theorem 5.2 (CM-v20-018: Windowed representative threshold composition). *Let $\tau, \sigma > 0$. If all relevant representatives exist, are window-normalized, and use the same monitored scope and window, then*

$$C_{\sigma,W}(C_{\tau,W}F) \simeq_{\text{PH}, I_{\text{mon}}} C_{\max\{\tau,\sigma\},W}F.$$

Proof. Use CM-v20-004 on each monitored profile. Window normalization prevents a second crop from creating new short bars. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-018
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-B-065
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-018; inherited at immutable v19 strength.

Theorem 5.3 (CM-v20-019: Canonical two-term Ext^1 -edge identification). *For $A \in D^{[-1,0]}(k\text{-mod})$, there is a canonical natural isomorphism*

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

Proof. Apply $\text{Hom}_{D^b}(-, k[1])$ to the canonical truncation triangle. Since k -vector spaces are semisimple, the outer Ext_k^1 and Ext_k^2 terms vanish, leaving the displayed natural isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-019
Role / status: core_micro_theorem / proved
Source claims: CR-CH4-016, CR-C-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5
Repair ancestry: CM-v19-019; inherited at immutable v19 strength.

Theorem 5.4 (CM-v20-020: Conditional one-way repaired PH_1 -to- Ext^1 bridge). *If $(F; W, \tau)$ has a complete eligible realization record and*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Eligibility places $A = \mathcal{R}(C_{\tau,W}F)$ in $D^{[-1,0]}(k\text{-mod})$ and supplies the edge-realization isomorphism between $H^{-1}(A)$ and the repaired degree-one residual edge. Vanishing of the repaired profile gives $H^{-1}(A) = 0$; CM-v20-019 then gives the Ext vanishing. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-020
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-025, CR-C-026
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5
Repair ancestry: CM-v19-020; inherited at immutable v19 strength.

Theorem 5.5 (CM-v20-021: Reverse Ext^1 -to-repaired-profile implication fails). *Under the standing eligibility axioms alone,*

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \not\Rightarrow \text{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Take a nonzero repaired interval profile and the constant-zero realization with constant-zero edge extractor. The amplitude and edge-realization requirements are satisfied at the zero extracted edge, and $\text{Ext}^1(0, k) = 0$, while the repaired profile remains nonzero. Reverse recovery therefore requires a separate faithfulness theorem. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-021
Role / status: core_micro_theorem / proved
Source claims: CR-C-038, CR-CH4-022; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-021; inherited at immutable v19 strength.

6 Terminal-generic Type IV micro-theorems

Policy 6.1 (CM-v20-022: Actual artifact and terminal tameness are prerequisites for Type IV). Type IV is assigned exactly one status:

`not_invoked, undefined, certified_zero, obstructed.`

Numerical diagnostics are defined only after an actual tower comparison morphism, typed source and target, terminal-tame kernel and cokernel profiles, fixed $(W, \tau, I_{\text{mon}})$, and artifact provenance are present. Missing or ill-typed input yields undefined, never $(0, 0)$.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-022
Role / status: operational_policy / operational
Source claims: CR-CH5-003, CR-CH5-014, CR-D-017, CR-D-023
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-022; inherited at immutable v19 strength.

Theorem 6.2 (CM-v20-023: Characterization of certified Type IV-free status). *A tower diagnostic record has status `certified_zero` if and only if its comparison artifact is valid, its kernel and cokernel defect profiles are terminal-tame, and*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

With the first two clauses valid, a nonzero pair gives `obstructed`.

Proof. These are exactly the validity, terminal-tameness, and numerical clauses defining the typed diagnostic record. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-023
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-019, CR-D-026
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-023; inherited at immutable v19 strength.

Theorem 6.3 (CM-v20-024: Zero diagnostics plus no interior defects imply isomorphism). *Assume the defect profiles are terminal-tame and interval-decomposable. If the terminal-generic diagnostics vanish and the kernel and cokernel contain no interior or noncofinal interval summands, then the tower comparison artifact is an isomorphism in the declared profile category.*

Proof. The zero diagnostics exclude terminal-cofinal summands. The no-interior-defect hypothesis excludes all remaining interval summands. Thus kernel and cokernel vanish. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-024
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-029, CR-D-029
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-024; inherited at immutable v19 strength.

Theorem 6.4 (CM-v20-025: Full-isomorphism band implies terminal-generic stable band). *Every artifact-coherent threshold band on which each monitored tower comparison artifact is an isomorphism is a global terminal-generic Type IV stable band.*

Proof. Every isomorphism has zero kernel and cokernel, hence terminal-tame zero diagnostics and a certified_zero record at every threshold, assuming the remaining artifact fields are present. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-025
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-031, CR-D-054
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-025; inherited at immutable v19 strength.

7 Restart, obligation, and countable-audit micro-theorems

Theorem 7.1 (CM-v20-026: Finite-horizon product-weighted Restart bound). *If transitions $0 \rightarrow 1, \dots, n-1 \rightarrow n$ satisfy*

$$r_{k+1} \geq \kappa_k r_k - \eta_k, \quad 0 < \kappa_k \leq 1,$$

then

$$r_n \geq \left(\prod_{j=0}^{n-1} \kappa_j \right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j \right) \eta_m.$$

Proof. Induct on n , substitute the previous bound into $r_{n+1} \geq \kappa_n r_n - \eta_n$, and distribute κ_n . $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-026
Role / status: core_micro_theorem / proved
Source claims: CR-CH6-032, CR-E-025
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-026; inherited at immutable v19 strength.

Theorem 7.2 (CM-v20-027: Conditional countable Restart-summability principle). *Set $K_0 = 1$ and $K_n = \prod_{j=0}^{n-1} \kappa_j$. Suppose the initial reserve is positive, every transition is certified, every target record is complete with all invoked non-reserve clauses passing, no required record is undefined, and*

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Then every finite-stage reserve is positive and the ordered record family is an auditable sequential certificate chain. A uniform positive lower bound additionally requires $\inf_n K_n > 0$.

Proof. CM-v20-026 gives

$$\frac{r_n}{K_n} \geq r_0 - \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}} > 0.$$

The artifact and target-record hypotheses give reconstructibility. Uniform positivity does not follow if $K_n \rightarrow 0$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-027
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH6-037, CR-E-028
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-027; inherited at immutable v19 strength.

Theorem 7.3 (CM-v20-028: No scalar manufacture of non-scalar evidence). *No increase in a metric budget, threshold gap, reserve, or summability allowance can discharge a missing representative, actual morphism, eligibility record, terminal-tameness certificate, higher-coherence record, Claim UID, manifest field, or artifact identity.*

Proof. These obligations are typed as non-scalar atoms. The scalar ledger and the non-scalar obligation registry have disjoint semantic codomains; no scalar inequality constructs an absent typed witness. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-028
Role / status: core_micro_theorem / proved
Source claims: CR-CH7-015, CR-F-004
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-028; inherited at immutable v19 strength.

Theorem 7.4 (CM-v20-029: Non-compensation and dependency blocker propagation). *In the typed verdict algebra, a passing sibling cannot cancel a rejected or required undefined atom. If a required dependency is not passing, every dependent theorem-evidence obligation is blocked from pass until that dependency is repaired.*

Proof. Certificate readiness is a typed conjunction over dependency-closed obligations. Conjunction is downward closed: pass requires every required atom and dependency to pass. Therefore neither numerical surplus nor an unrelated passing atom can compensate for a blocker. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-029
Role / status: core_micro_theorem / proved
Source claims: CR-F-026, CR-F-030, CR-F-061
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-029; inherited at immutable v19 strength.

Theorem 7.5 (CM-v20-030: Finite prefixes do not certify countable readiness). *A finite prefix of a countable Restart sequence, overlap atlas, registry, or manifest cannot certify the complete countable object unless a separate theorem reduces the infinite obligation to finite data.*

Proof. Any finite prefix omits later entries and their dependencies. Without a reduction theorem, omitted required obligations remain unchecked, so complete readiness is undefined rather than passed. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-030
Role / status: core_micro_theorem / proved
Source claims: CR-F-045, CR-GAPP-012
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-030; inherited at immutable v19 strength.

8 Bridge-ready internal theorem

Theorem 8.1 (CM-v20-031: Eventually constant repaired tower with apex agreement). *Let*

$$Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow \dots$$

be a repaired profile tower in a category admitting the declared colimit. Suppose there exists N such that every transition after N is an isomorphism, the apex profile Q_∞ is equipped with a declared isomorphism to the stabilized profile Q_N , and the canonical colimit-to-apex artifact is the induced comparison. Then that artifact is an isomorphism. If the Type IV input is terminal-tame and valid, its diagnostics are $(0, 0)$.

Proof. The colimit of an eventually constant diagram is canonically isomorphic to its stabilized value. Composing with the declared apex agreement gives an isomorphism. Its kernel and cokernel vanish, so the terminal-generic diagnostics vanish. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-031
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH8-012, CR-CH8-013, CR-D-047, CR-D-048
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-031; inherited at immutable v19 strength.

Remark 8.2 (Bridge boundary). CM-v20-031 is an internal bridge-ready theorem. It does not prove that any arithmetic, PDE, geometric, categorical, or cryptographic tower realizes into such a diagram. That transport is a separate Interface or Bridge Theorem.

9 Exact-invariance and terminal-generic defect calculus

Theorem 9.1 (CM-v20-032: Representative-invariance of eligible Ext^1 discharge). *Let $C_{\tau,WF}$ and $C'_{\tau,WF}$ be admissible representatives with eligible realizations*

$$A := \mathcal{R}(C_{\tau,WF}), \quad A' := \mathcal{R}(C'_{\tau,WF})$$

in $D^{[-1,0]}(k\text{-mod})$. Suppose an isomorphism $\eta : A \xrightarrow{\sim} A'$ is compatible with the declared degree-one edge-realization records. Then precomposition induces an isomorphism

$$\eta^* : \text{Ext}^1(A', k) \xrightarrow{\sim} \text{Ext}^1(A, k),$$

and consequently

$$\text{Ext}^1(A, k) = 0 \iff \text{Ext}^1(A', k) = 0.$$

Thus an eligible categorical discharge is invariant under an artifact-backed change of realization-equivalent representative.

Proof. The contravariant functor $\text{Hom}_{D(k)}(-, k[1])$ sends isomorphisms to isomorphisms. Compatibility of the edge records ensures that the two Ext groups discharge the same declared repaired degree-one clause rather than merely two unrelated derived objects. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-032
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-036, CR-B-057
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-032; inherited at immutable v19 strength.

Theorem 9.2 (CM-v20-033: Exact-conjugacy invariance of Type IV diagnostics). *Let $\phi : M \rightarrow N$ and $\phi' : M' \rightarrow N'$ be actual comparison morphisms and suppose there are isomorphisms $\alpha : M \xrightarrow{\sim} M'$, $\beta : N \xrightarrow{\sim} N'$ such that*

$$\beta \circ \phi = \phi' \circ \alpha.$$

Then

$$\ker \phi \cong \ker \phi', \quad \text{coker } \phi \cong \text{coker } \phi'.$$

Whenever both records are terminal-tame on the same monitored scope,

$$\mu(\phi) = \mu(\phi'), \quad \nu(\phi) = \nu(\phi'),$$

and the corresponding total collapse diagnostics agree degreewise and after aggregation.

Proof. The isomorphism α restricts to the kernels, and β descends to the cokernels. Terminal-generic dimension is invariant under persistence-module isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-033
Role / status: core_micro_theorem / proved
Source claims: CR-D-062;CM-localproof
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING,FR-T3,FR-T5
Repair ancestry: CM-v19-033; inherited at immutable v19 strength.

Theorem 9.3 (CM-v20-034: Terminal-generic index identity). *Let $\phi : M \rightarrow N$ be an actual morphism between terminal-tame persistence modules with finite-dimensional stabilized tails. Then*

$$\mu(\phi) - \nu(\phi) = \text{tgdim}(M) - \text{tgdim}(N).$$

Proof. Choose a common terminal parameter after which M , N , and the defect profiles are constant up to the declared tail identifications. The induced finite-dimensional map $\phi_\infty : M_\infty \rightarrow N_\infty$ satisfies

$$\dim \ker \phi_\infty - \dim \text{coker } \phi_\infty = \dim M_\infty - \dim N_\infty$$

by rank-nullity. These four dimensions are precisely the declared terminal-generic dimensions. $\square$

Corollary 9.4 (Balanced-tail consequence). *If $\text{tgdim}(M) = \text{tgdim}(N)$, then $\mu(\phi) = \nu(\phi)$. In particular, under the valid terminal-tame record, either one vanishes if and only if the other vanishes.*

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-034
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-011,CR-D-013;CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING,FR-T4,FR-T5
Repair ancestry: CM-v19-034; inherited at immutable v19 strength.

Theorem 9.5 (CM-v20-035: Direct-sum additivity of Type IV diagnostics). *For terminal-tame actual morphisms $\phi_1 : M_1 \rightarrow N_1$ and $\phi_2 : M_2 \rightarrow N_2$,*

$$\mu(\phi_1 \oplus \phi_2) = \mu(\phi_1) + \mu(\phi_2), \quad \nu(\phi_1 \oplus \phi_2) = \nu(\phi_1) + \nu(\phi_2).$$

The same statement holds for every finite direct sum and for the declared finite aggregation over monitored degrees.

Proof. Kernels and cokernels commute with finite biproducts in the persistence-module abelian category:

$$\ker(\phi_1 \oplus \phi_2) \cong \ker \phi_1 \oplus \ker \phi_2, \quad \text{coker}(\phi_1 \oplus \phi_2) \cong \text{coker } \phi_1 \oplus \text{coker } \phi_2.$$

Terminal-generic dimension is additive on finite direct sums. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-035
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-015, CR-D-020; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-035; inherited at immutable v19 strength.

Theorem 9.6 (CM-v20-036: Composition subadditivity of Type IV diagnostics). *Let*

$$M \xrightarrow{f} N \xrightarrow{g} P$$

be terminal-tame actual morphisms on a common terminal scope. Then

$$\mu(gf) \leq \mu(f) + \mu(g), \quad \nu(gf) \leq \nu(f) + \nu(g).$$

Consequently, two valid terminal-generically clean comparisons have a clean composite.

Proof. On a common stabilized tail these are finite-dimensional linear maps. The map $\ker(gf) \rightarrow \ker g$, $x \mapsto f(x)$, has kernel $\ker f$, giving the kernel inequality. Also

$$\text{rank}(gf) \geq \text{rank}(f) + \text{rank}(g) - \dim N,$$

which rearranges to the cokernel inequality. Tail dimensions equal the terminal-generic diagnostics. $\square$

Remark 9.7 (No converse). Cleanliness of gf does not force cleanliness of f or g ; defects may be killed or bypassed by composition.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-036
Role / status: core_micro_theorem / proved
Source claims: CR-D-023; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-036; inherited at immutable v19 strength.

Theorem 9.8 (CM-v20-037: Snake-lemma control over short exact comparison diagrams). *Consider a commutative diagram of persistence modules with exact rows*

$$\begin{array}{ccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' \rightarrow 0 \\ & & \downarrow \phi' & & \downarrow \phi & & \downarrow \phi'' \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' \rightarrow 0. \end{array}$$

Assume the six modules and the vertical defect profiles are terminal-tame on a common terminal scope. Then

$$\mu(\phi) \leq \mu(\phi') + \mu(\phi''), \quad \nu(\phi) \leq \nu(\phi') + \nu(\phi'').$$

In particular, terminal-generic cleanliness of both outer vertical maps implies cleanliness of the middle map.

Proof. Pointwise exactness and the snake lemma give

$$0 \rightarrow \ker \phi' \rightarrow \ker \phi \rightarrow \ker \phi'' \rightarrow \text{coker } \phi' \rightarrow \text{coker } \phi \rightarrow \text{coker } \phi'' \rightarrow 0.$$

Pass to a common stabilized tail and take finite-dimensional vector-space dimensions. Exactness bounds the middle kernel and cokernel dimensions by the sums of their outer counterparts. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-037
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-023; CM-localproofviasnakelemma
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-037; inherited at immutable v19 strength.

10 Single-morphism metric bridge

Definition 10.1 (Finite-shift triviality). For $\epsilon \geq 0$, a persistence module E is ϵ -trivial if its translation morphism $E \rightarrow E(\epsilon)$ is zero.

Theorem 10.2 (CM-v20-038: Single-morphism characterization of one-parameter interleaving). *Let M, N be pointwise finite-dimensional one-parameter persistence modules and let $\delta \geq 0$. Then M and N are δ -interleaved if and only if there exists a morphism*

$$f : M \longrightarrow N(\delta)$$

whose kernel and cokernel are both 2δ -trivial.

Proof. This is Corollary 6.6 of Bauer and Lesnick (2015). Its one-parameter and pointwise finite-dimensional hypotheses are part of the theorem and are not silently weakened. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-038
Role / status: core_micro_theorem / proved
Source binding: Bauer--Lesnick2015, Corollary6.6;CM-localimportproofrecord
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-038; inherited at immutable v19 strength.

Theorem 10.3 (CM-v20-039: Finite-trivial defect implies terminal-generic cleanliness). *Let $f : M \rightarrow N$ be an actual morphism whose kernel and cokernel are interval-decomposable, terminal-tame, and ϵ -trivial for some finite ϵ . Then*

$$\mu(f) = \nu(f) = 0.$$

Consequently, a morphism satisfying the defect hypothesis of CM-v20-038 is terminal-generically clean whenever it is used as the declared Type IV comparison artifact on the same scope.

Proof. A terminal-cofinal interval summand has nonzero structure maps across every finite shift sufficiently far out. Such a summand cannot occur in an ϵ -trivial module. Hence neither defect profile has a terminal-cofinal summand, so both terminal-generic dimensions vanish. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-039
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-056;Bauer--Lesnick2015;CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-039; inherited at immutable v19 strength.

Theorem 10.4 (CM-v20-040: Terminal-generic cleanliness does not imply quantitative interleaving). *For every $\delta > 0$, terminal-generic zero of an actual comparison morphism does not imply that its kernel and cokernel are 2δ -trivial, and does not imply a δ -interleaving.*

Proof. Choose $L > 2\delta$, let $M = 0$, $N = k_{[0,L]}$, and take the zero morphism $f : M \rightarrow N(\delta)$. Its kernel is zero and its cokernel is a finite interval module, so both terminal-generic diagnostics vanish. However, the cokernel is not 2δ -trivial because its bar length is greater than 2δ . Moreover

$$d_I(M, N) = d_B(\emptyset, \{[0, L]\}) = L/2 > \delta.$$

Thus Type IV cleanliness is strictly weaker than a bounded-defect interleaving certificate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-040
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-027, CR-D-038; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-040; inherited at immutable v19 strength.

11 v20 repaired-zero and detector-transport micro-theorems

Policy 11.1 (CM-v20-041: Detector stage and completeness boundary). Every detector theorem is consumed only at its registered profile stage. Nonzero soundness and zero completeness are independent. A zero detector value is theorem evidence only when the applicable endpoint, birth or germ exhaustiveness, stencil or mesh coverage, family membership, excess-source, and artifact obligations are complete; otherwise the mathematical value is undefined.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-041
Role / status: operational_policy / operational
Source claims: CR-G-003, CR-G-004, CR-CH1-006, CR-CH1-007, CR-CH1-008, CR-CH1-009
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-NONVERDICT, FR-T5
Repair ancestry: newv20detector-governancemicro-policy

Theorem 11.2 (CM-v20-042: Distance to zero of a finite repaired profile). *Let P be a finite barcode profile. If P contains an infinite bar, then $d_B(P, 0) = +\infty$. If all bars are finite, then*

$$d_B(P, 0) = \frac{1}{2} \max_{I \in P} \ell(I),$$

with $d_B(0, 0) = 0$. In particular, if P is τ -reduced and nonzero, then $d_B(P, 0) > \tau/2$.

Proof. Against the zero barcode every finite bar must be matched to the diagonal, with cost $\ell(I)/2$, while an infinite bar has no finite diagonal match. The bottleneck cost is therefore the displayed maximum. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-042
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-018, CR-CH2-019, CR-A-019, CR-A-020
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 11.3 (CM-v20-043: Sharp zero separation for τ -reduced profiles). *Let P, Q be finite τ -reduced profiles and suppose $d_B(P, Q) \leq \varepsilon$ with $2\varepsilon \leq \tau$. Then*

$$P = 0 \iff Q = 0.$$

The constant is sharp: for $2\varepsilon > \tau$ the equivalence can fail.

Proof. If $P = 0$ and $Q \neq 0$, then an infinite bar in Q forces infinite distance, while a finite retained bar has length $> \tau \geq 2\varepsilon$ and hence distance to zero $> \varepsilon$ by CM-v20-042. The converse is symmetric. For sharpness choose one bar of length L with $\tau < L \leq 2\varepsilon$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-043
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-020, CR-A-021
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 11.4 (CM-v20-044: Metric-safe repaired-zero transport). *Let B, C be finite barcode profiles and suppose $d_B(B, C) \leq \varepsilon$. Assume that no finite bar of either profile has length in $[\tau - \varepsilon, \tau + \varepsilon]$ and that $2\varepsilon \leq \tau$. Then*

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

The operational certificate uses the strict reserves $\varepsilon < \text{gap}_\tau(B, C)$ and $2\varepsilon < \tau$.

Proof. CM-v20-012 transports the bottleneck bound through hard deletion. Both outputs are τ -reduced, so CM-v20-043 transports the exact zero predicate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-044
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH2-021, CR-CH2-022, CR-A-023, CR-A-024
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Corollary 11.5 (CM-v20-045: Detector-certified metric zero transport). *Assume a complete detector certificate proves $T_\tau^{\text{bar}}(B) = 0$. If the hypotheses of CM-v20-044 hold for (B, C) , then $T_\tau^{\text{bar}}(C) = 0$. Only the certified zero predicate is transported; detector ranks, sample locations, birth sets, germ sets, and certificate size are not transported.*

Proof. Apply CM-v20-044 to the source zero statement supplied by the complete detector certificate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-045
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH2-024, CR-CH2-025, CR-A-026, CR-A-027
Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 11.6 (CM-v20-046: Raw pointwise detector ranks are not bottleneck-stable). *There is no general implication*

$$d_B(B, C) \leq \varepsilon \implies \dim \text{im}(B(t) \rightarrow B(t+s)) = \dim \text{im}(C(t) \rightarrow C(t+s))$$

for fixed t, s , even for one-bar finite barcodes and arbitrarily small positive ε .

Proof. Translate one endpoint of a one-bar interval by an arbitrarily small δ across the fixed sampling point while keeping the span map nonzero on one profile and zero on the other. The bottleneck distance is at most δ , but the ranks differ. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-046
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-026, CR-CH2-027, CR-A-028, CR-A-029
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

12 Finite detector certificates and proof compression

Theorem 12.1 (CM-v20-047: Finite-stencil nonzero soundness). *Let B be a constructible windowed profile, let T be a finite subset of the effective τ -domain, and set*

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \rightarrow B(t + \tau)).$$

Then

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \implies T_{\tau}^{\text{bar}}(B) \neq 0.$$

No coverage hypothesis is required for this direction.

Proof. A nonzero image component is supported by an interval summand containing both t and $t + \tau$. That summand has window-relative length $> \tau$ and is retained. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-047

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-025, CR-CH3-027, CR-CH3-028, CR-B-014, CR-B-016, CR-B-017

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.2 (CM-v20-048: Birth detector zero-reflection). *For a finite left-closed right-open windowed barcode B with a certified exhaustive birth set $T_{\text{birth},\tau}(B)$,*

$$D_{T_{\text{birth},\tau}(B),\tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. Soundness is CM-v20-047. Every retained interval $[p, q)$ satisfies $p + \tau < q$, so the structure map at its certified birth p is nonzero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-048

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-029, CR-CH3-030, CR-B-018, CR-B-019

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.3 (CM-v20-049: Decoration-invariant germ detector zero-reflection). *For a finite decorated windowed barcode with a certified exhaustive geometric-birth set, the finite direct sum of shifted right-germ images is zero if and only if the repaired profile is zero. This statement is independent of finite-endpoint open/closed decoration.*

Proof. For an interval with geometric endpoints $p < q$, the shifted right-germ map is nonzero exactly when $p + \tau < q$. Equality gives zero because the target germ is taken strictly to the right. Summing over the exhaustive geometric-birth set yields the equivalence. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-049

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-031, CR-CH3-032, CR-CH3-033, CR-B-020, CR-B-021, CR-B-022

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.4 (CM-v20-050: Coverage-certified finite-stencil completeness). *Let T be a finite stencil. If an artifact-backed coverage certificate proves*

$$\forall I \in \text{Ret}_{\tau, W}(B), \quad T \cap A_{\tau}(I) \neq \emptyset,$$

then

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. CM-v20-047 gives one direction. If the repaired profile is nonzero, coverage supplies a sample in the active-start region of a retained interval, making the stencil nonzero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-050
Role / status: core_micro_theorem / proved
Source claims: CR-CH3-034, CR-CH3-035, CR-B-023, CR-B-024
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 12.5 (CM-v20-051: Family-mesh detector completeness). *Let $\mathcal{F}$ be a declared finite-barcode family on a bounded window and suppose an independently proved uniform retained-bar excess lower bound $e_{\tau, W}(\mathcal{F}) > 0$ is available. If a finite stencil has augmented mesh strictly smaller than this bound, then it is complete for every $B \in \mathcal{F}$:*

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. Every retained interval has an active-start interval of length at least the certified excess. The strict augmented-mesh inequality forces the stencil to meet each such interval. Apply CM-v20-050. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-051
Role / status: core_micro_theorem / proved
Source claims: CR-CH3-036, CR-CH3-037, CR-CH3-038, CR-CH3-039, CR-B-025, CR-B-026, CR-B-027, CR-B-028
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 12.6 (CM-v20-052: Exact augmented-mesh stencil count). *Let the effective detector interval have length $L_{\text{eff}} \geq 0$ and let $\gamma > 0$. The minimum number of interior stencil points required to make every augmented gap at most γ is*

$$m_{\text{mesh}}(L_{\text{eff}}, \gamma) = \max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{\gamma} \right\rceil - 1 \right\}.$$

Proof. With $N = \lceil L_{\text{eff}}/\gamma \rceil$, $N - 1$ equally spaced interior points produce N gaps of size at most γ . Conversely, m points produce $m + 1$ augmented gaps whose lengths sum to L_{eff} , forcing $L_{\text{eff}} \leq (m + 1)\gamma$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-052
Role / status: core_micro_theorem / proved
Source claims: CR-CH3-050, CR-CH3-051, CR-B-040, CR-B-041
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

Corollary 12.7 (CM-v20-053: Family-relative MinCert upper bounds). *For a detector information class Θ :*

(i) *a finite certified union of exhaustive birth sets gives $\text{MinCert}(\Theta)$ at most the size of that union;*

(ii) under CM-v20-051, every $0 < \gamma < e_{\tau, W}(\mathcal{F})$ gives

$$\text{MinCert}(\Theta) \leq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

These are family-relative upper bounds, not retrospective objectwise compression claims.

Proof. The birth-set union is complete by CM-v20-048. The mesh bound follows from CM-v20-051 and CM-v20-052. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-053

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-047, CR-CH3-048, CR-CH3-049, CR-CH3-052, CR-B-037, CR-B-038, CR-B-039, CR-B-042

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.8 (CM-v20-054: No uniform finite detector on an unbounded birth family). *On $W = [a, +\infty)$, fix $\tau, e > 0$ and consider the translation family*

$$\mathcal{U}_{a, \tau, e} = \{[p, p + \tau + e) : p \geq a\}.$$

For every finite stencil T there exists a member whose stencil value is zero but whose repaired profile is nonzero. Consequently the corresponding family-relative MinCert is $+\infty$.

Proof. A finite stencil is bounded above. Choose p larger than every sample. The retained bar has active-start region $[p, p + e)$, disjoint from the stencil. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-054

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-053, CR-B-043

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.9 (CM-v20-055: Exact-stage detector invariance). *An actual persistence-profile isomorphism on the declared window induces isomorphisms of all fixed-stencil image spaces and, under the same geometric-birth convention, of the right-germ detectors. Detector zero/nonzero is therefore invariant under exact source-stage isomorphism.*

Proof. Naturality gives a commutative square for each structure map; vertical isomorphisms induce isomorphisms on images. Finite direct sums and right-germ colimits preserve the resulting isomorphisms. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-055

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH3-043, CR-B-032, CR-B-059

Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.10 (CM-v20-056: Metric zero-verdict composition). *Suppose complete detectors zero-reflect $T_{\tau}^{\text{bar}}(B)$ and $T_{\tau}^{\text{bar}}(C)$, and the profiles satisfy the two-sided threshold-gap hypotheses with $d_B(B, C) \leq \varepsilon$ and $2\varepsilon \leq \tau$. Then the two detector values are simultaneously zero. The operational certificate uses $2\varepsilon < \tau$.*

Proof. Compose the two detector zero-reflection equivalences with CM-v20-044. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-056
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH3-044, CR-B-033, CR-B-034
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

13 Higher obstruction edge micro-theorems

Theorem 13.1 (CM-v20-057: Canonical higher field-coefficient Ext edge). *Let k be a field, $A \in D^b(k\text{-mod})$, and $n \geq 0$. There is a canonical isomorphism, natural contravariantly in A ,*

$$\text{Ext}^n(A, k) \cong \text{Hom}_k(H^{-n}(A), k).$$

Consequently $\text{Ext}^n(A, k) = 0$ if and only if $H^{-n}(A) = 0$.

Proof. The hyper-Ext spectral sequence has $E_2^{p,q} = \text{Ext}_k^p(H^{-q}(A), k)$. Semisimplicity of finite-dimensional vector spaces over a field kills all columns $p > 0$, leaving the stated natural edge isomorphism. Linear duality reflects zero in finite dimension. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-057
Role / status: core_micro_theorem / proved
Source claims: CR-CH4-013, CR-C-005, CR-C-006, CR-C-007, CR-C-008
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.2 (CM-v20-058: Standard n -eligible higher Ext discharge). *Let $n \geq 1$ and suppose $(F; W, \tau)$ has a complete standard n -eligible realization record with edge extractor E_n and an artifact-backed identification*

$$H^{-n}(A_{n;F,W,\tau}) \cong E_n(\text{Eval}_{n,W,\tau}(F)).$$

If $E_n(0) = 0$, then

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. The source zero gives $E_n(\text{Eval}) = 0$. Apply the edge-realization identification and CM-v20-057. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-058
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-019, CR-CH4-020, CR-C-022, CR-C-023, CR-C-024
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.3 (CM-v20-059: Natural zero-reflection upgrades discharge to equivalence). *Under the hypotheses of CM-v20-058, assume the repaired profile lies in a declared class $\mathcal{P}_n$ and a separately proved natural zero-reflection package establishes*

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n).$$

Then

$$\text{Eval}_{n,W,\tau}(F) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. The forward direction is CM-v20-058. Conversely, Ext vanishing and CM-v20-057 imply $H^{-n}(A) = 0$; the edge-realization certificate gives $E_n(P) = 0$, and the independent zero-reflection theorem gives $P = 0$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-059
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH1-012, CR-CH4-021, CR-CH4-022, CR-C-027, CR-C-028, CR-C-029, CR-C-037
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.4 (CM-v20-060: Synthetic duality does not create a natural bridge). *A shifted detector vector space or toy-derived object may satisfy an algebraic identity*

$$\text{Ext}^n(A^{\text{syn}}, k) \cong \text{Hom}_k(D(P), k),$$

but this does not establish a natural realization of P , representative independence, a PH_n -to- Ext^n theorem, or an external-domain bridge.

Proof. The displayed identity concerns the chosen synthetic object. The missing realization, naturality, zero-reflection, representative, and external Return arrows are independent typed obligations. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-060
Role / status: core_micro_theorem / proved
Source claims: CR-G-007, CR-CH4-039, CR-CH4-040, CR-CH4-041, CR-C-034, CR-C-035, CR-C-036
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.5 (CM-v20-061: Representative invariance of higher Ext discharge). *If two standard n -eligible representatives are connected by a recorded realization isomorphism compatible with their higher edge-realization certificates, then their Ext groups are identified and the truth value of the higher Ext-vanishing clause is independent of the representative choice.*

Proof. The contravariant derived Hom functor carries the realization isomorphism to an Ext isomorphism. Compatibility with the edge certificates identifies the complete discharge data. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-061
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-033, CR-CH4-034, CR-CH4-036, CR-B-057
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.6 (CM-v20-062: Complete detector higher Ext discharge). *Let $n \geq 1$, let $(F; W, \tau)$ be standard n -eligible, and let a complete detector certificate on the exact repaired-profile stage establish that detector zero implies $\text{Eval}_{n,W,\tau}(F) = 0$. A certified zero detector value then implies*

$$\text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

If the detector is also the declared natural zero-reflecting edge, detector zero is equivalent to Ext vanishing on the registered class.

Proof. Detector completeness gives repaired-profile zero; CM-v20-058 gives Ext zero. The equivalence clause is CM-v20-059 specialized to the declared detector edge. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-062
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-024, CR-CH4-037, CR-C-032, CR-C-033
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

14 Terminal Type IV, transient defect, and full-isomorphism calculus

Theorem 14.1 (CM-v20-063: Terminal and interior defect separation). *For an interval-decomposable terminal-tame profile M on W , let $\text{Cof}_W(M)$ be the cofinal subprofile and $\text{Int}_W(M)$ the noncofinal subprofile. Then*

$$M \cong \text{Cof}_W(M) \oplus \text{Int}_W(M), \quad \text{tgdim}_W(M) = \#\mathcal{B}(\text{Cof}_W(M)),$$

with multiplicity. Hence $\text{tgdim}_W(M) = 0$ if and only if $M \cong \text{Int}_W(M)$.

Proof. The terminal-generic value counts exactly the terminal-cofinal interval summands. The barcode partitions canonically into cofinal and noncofinal summands. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-063
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-021, CR-D-031
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.2 (CM-v20-064: Complete transient-defect detector characterization). *Let ϕ be a valid tower comparison morphism with constructible kernel and cokernel profiles. Rebuild complete detector certificates at source stages `kernel_defect` and `cokernel_defect`, with a declared defect threshold and either the full or interior component. Then detector zero is equivalent to zero of the corresponding hard-thresholded defect profile, and detector nonzero is equivalent to a retained defect bar.*

Proof. The interval-image, birth, germ, coverage, and mesh detector theorems apply to the supplied defect profiles after stage-specific reconstruction. No pre-threshold detector artifact is relabeled. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-064
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-022, CR-CH5-023, CR-CH5-024, CR-D-033, CR-D-034, CR-D-035
Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared
Failure routes: FR-MISSING, FR-TRANSIENT, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.3 (CM-v20-065: Joint terminal and long-transient classification). *Assume a valid comparison artifact, terminal-tame defect profiles, and complete interior defect detectors. Degreewise, and after finite monitored aggregation:*

- (i) *nonzero (μ, ν) certifies terminal/cofinal Type IV, independently of any interior defect;*
- (ii) *$(\mu, \nu) = (0, 0)$ with a certified nonzero interior detector gives a long noncofinal transient defect;*

- (iii) $(\mu, \nu) = (0, 0)$ with certified-zero interior detectors excludes terminal defects and interior bars longer than the defect threshold, but not shorter defects.

Proof. CM-v20-063 classifies cofinal defects and CM-v20-064 classifies retained interior defects. Their conjunctions give the cases without compensation. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-065
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-026, CR-CH5-027, CR-D-037, CR-D-038
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-TRANSIENT, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.4 (CM-v20-066: Conditional full-isomorphism upgrade). *Let ϕ be a valid morphism in the declared abelian profile category. Suppose terminal diagnostics vanish, complete interior defect detectors are certified zero at τ_{def} , and a separate short-defect exclusion certificate rules out every remaining nonzero interior bar of length at most τ_{def} . Then*

$$\ker \phi = 0, \quad \text{coker } \phi = 0,$$

so ϕ is an isomorphism.

Proof. Terminal zero excludes cofinal defect bars. Detector zero excludes interior bars longer than the defect threshold. The short-defect theorem excludes the remainder. A morphism in an abelian category with zero kernel and cokernel is an isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-066
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-028, CR-CH5-029, CR-D-039, CR-D-040
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.5 (CM-v20-067: Terminal cleanliness is strictly weaker than full or metric cleanliness). *Vanishing terminal-generic diagnostics does not imply full isomorphism, absence of finite transient defects, a quantitative interleaving bound, or finite-shift triviality of the full kernel and cokernel profiles.*

Proof. A zero morphism $0 \rightarrow k_{[0,L]}$ has zero terminal diagnostics for finite L but nonzero cokernel; choosing L arbitrarily large defeats any prescribed interleaving or finite-shift bound. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-067
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-027, CR-CH5-029, CR-D-038, CR-D-040; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

15 Detector-aware Restart, atlas coverage, and finite-to-infinite reduction

Theorem 15.1 (CM-v20-068: Coverage requirement for detector atlases). *Let a global repaired-profile interface and a finite or countable family of complete local detector certificates be given. Local detector-zero*

verdicts imply global repaired-profile zero only when a separately verified coverage record proves that every global retained bar is visible in at least one local detector scope. Pairwise overlap coherence alone is insufficient.

Proof. Without coverage a retained bar may cross local boundaries or lie outside all active-start regions while every local detector is zero. With coverage, any global retained bar would create a nonzero local detector, contradicting local zero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-068
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH6-026, CR-E-049, CR-E-050, CR-E-051, CR-E-052
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.2 (CM-v20-069: Detector-aware Restart handoff is nonautomatic). *A certified Restart reserve inequality does not transport a detector-zero predicate across a window transition. A valid handoff additionally requires a target-zero route and boundary-bridge coverage sufficient to expose every retained bar crossing the partition boundary.*

Proof. Restart controls normalized reserve loss, not detector completeness or bar visibility. The missing target-zero and boundary-coverage clauses are non-scalar obligations. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-069
Role / status: core_micro_theorem / proved
Source claims: CR-CH6-039, CR-CH6-040, CR-E-053, CR-E-054, CR-E-055
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.3 (CM-v20-070: Eventually constant exact finite-to-infinite reduction). *Let $M_0 \rightarrow M_1 \rightarrow \dots$ be a directed sequence with declared colimit L . Suppose all transitions are isomorphisms from some stage N onward. Then the canonical map $M_N \rightarrow L$ is an isomorphism. If a compatible apex cocone has $M_N \rightarrow A_\infty$ an isomorphism, the induced comparison $L \rightarrow A_\infty$ is an isomorphism; in an abelian profile category its kernel and cokernel vanish, all terminal diagnostics vanish, and every complete full or interior defect detector has zero value.*

Proof. The stabilized value supplies a cocone inverse to the colimit map by the universal property. Apex agreement factors as the composite of two isomorphisms. The defect conclusions follow from zero kernel and cokernel. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-070
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH8-012, CR-CH8-013, CR-D-046, CR-D-047, CR-D-048
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.4 (CM-v20-071: Finite prefixes do not determine an infinite reduction). *No finite prefix of a directed sequence determines its colimit, eventual constancy, apex agreement, or finite-to-infinite comparison defect without a separate reduction theorem and complete tail evidence.*

Proof. Two systems may agree through any prescribed finite stage, while one remains zero and the other becomes a nonzero constant system at the next stage. Their colimits differ. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-071
Role / status: core_micro_theorem / proved
Source claims: CR-CH8-020, CR-F-045, CR-GAPP-012
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.5 (CM-v20-072: Layered non-compensation for v20 certificates). *A passing detector layer cannot compensate for missing higher-edge eligibility; a passing higher-edge layer cannot compensate for an incomplete finite-to-infinite reduction; clean terminal diagnostics cannot compensate for transient-defect uncertainty; and artifact closure cannot compensate for a false theorem hypothesis. Every required layer and dependency must pass independently.*

Proof. The v20 readiness object is a dependency-closed typed conjunction over detector, profile, derived, tower/defect, reduction/Return, and governance layers. Conjunction admits no cancellation across semantic sorts. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-072
Role / status: core_micro_theorem / proved
Source claims: CR-F-030, CR-F-060, CR-F-061, CR-F-069, CR-GAPP-048
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: new v20 micro-theorem.

16 CM register closure and migration

Policy 16.1 (CM-v20-073: CM-v20 register closure and source-binding discipline). A CM-v20 package is register-closed only when every CM UID is unique and stable, every source Claim UID or primary-source locator resolves, every conditional hypothesis is displayed, the comparison policy is compatible with its runtime mode, detector stage and completeness sources are explicit, failure routes are typed, human-readable and machine-readable registers agree, and no deprecated, synthetic, operational, or non-claim item is consumed as theorem evidence. Final promotion into CR-v20 requires a separate immutable promotion receipt.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-073
Role / status: operational_policy / operational
Source claims: CR-G-002, CR-G-014, CR-G-015, CR-G-016, CR-G-017, CR-F-070, CR-GAPP-038, CR-GAPP-039
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: supersedes CM-v19-041 closure record.

16.1 Canonical CM-v20 register summary

CM ID	Title	Status	Comparison policy
CM-v20-001	Core micro-theorem boundary and no external upgrade	operational	not_applicable
CM-v20-002	Infinite-bar threshold invariance	proved	not_applicable

CM ID	Title	Status	Comparison policy
CM-v20-003	Hard threshold deletion idempotence	proved	not_applicable
CM-v20-004	Hard threshold composition	proved	not_applicable
CM-v20-005	Retained-profile threshold monotonicity	proved	not_applicable
CM-v20-006	Positive-threshold short bars are not Serre	proved	not_applicable
CM-v20-007	Nested window restriction identity	proved	not_applicable
CM-v20-008	Fixed window-first repaired evaluation	operational	not_applicable
CM-v20-009	No unconditional crop-collapse commutation	proved	not_applicable
CM-v20-010	Exact crop-collapse criterion	conditional	fixed_exact
CM-v20-011	Global non-expansiveness of hard deletion fails	proved	not_applicable
CM-v20-012	Sharp two-sided threshold-gap-safe stability	proved	fixed_metric_gap _certified
CM-v20-013	Scalarized-ledger-safe threshold classification	conditional	fixed_metric_gap _certified
CM-v20-014	Exact comparison is not a positive-radius robustness certificate	proved	fixed_exact
CM-v20-015	Runtime comparison-mode trichotomy	operational	runtime_resolved
CM-v20-016	Overlap transfer under a scalarized threshold-gap certificate	conditional	fixed_metric_gap _certified
CM-v20-017	Windowed representative idempotence	conditional	fixed_exact
CM-v20-018	Windowed representative threshold composition	conditional	fixed_exact
CM-v20-019	Canonical two-term Ext^1 -edge identification	proved	not_applicable
CM-v20-020	Conditional one-way repaired PH_1 -to- Ext^1 bridge	conditional	not_applicable
CM-v20-021	Reverse Ext^1 -to-repaired-profile implication fails	proved	not_applicable
CM-v20-022	Actual artifact and terminal tameness are prerequisites for Type IV	operational	not_applicable
CM-v20-023	Characterization of certified Type IV-free status	proved	not_applicable
CM-v20-024	Zero diagnostics plus no interior defects imply isomorphism	conditional	not_applicable
CM-v20-025	Full-isomorphism band implies terminal-generic stable band	conditional	not_applicable
CM-v20-026	Finite-horizon product-weighted Restart bound	proved	not_applicable
CM-v20-027	Conditional countable Restart-summability principle	conditional	not_applicable
CM-v20-028	No scalar manufacture of non-scalar evidence	proved	not_applicable
CM-v20-029	Non-compensation and dependency blocker propagation	proved	not_applicable
CM-v20-030	Finite prefixes do not certify countable readiness	proved	not_applicable
CM-v20-031	Eventually constant repaired tower with apex agreement	conditional	fixed_exact
CM-v20-032	Representative-invariance of eligible Ext^1 discharge	conditional	fixed_exact
CM-v20-033	Exact-conjugacy invariance of Type IV diagnostics	proved	fixed_exact
CM-v20-034	Terminal-generic index identity	proved	not_applicable
CM-v20-035	Direct-sum additivity of Type IV diagnostics	proved	not_applicable
CM-v20-036	Composition subadditivity of Type IV diagnostics	proved	not_applicable
CM-v20-037	Snake-lemma control over short exact comparison diagrams	conditional	not_applicable
CM-v20-038	Single-morphism characterization of one-parameter interleaving	proved	not_applicable
CM-v20-039	Finite-trivial defect implies terminal-generic cleanliness	conditional	not_applicable
CM-v20-040	Terminal-generic cleanliness does not imply quantitative interleaving	proved	not_applicable
CM-v20-041	Detector stage and completeness boundary	operational	not_applicable
CM-v20-042	Distance to zero of a finite repaired profile	proved	not_applicable
CM-v20-043	Sharp zero separation for τ -reduced profiles	proved	fixed_metric_gap _certified
CM-v20-044	Metric-safe repaired-zero transport	conditional	fixed_metric_gap _certified

CM ID	Title	Status	Comparison policy
CM-v20-045	Detector-certified metric zero transport	conditional	runtime_resolved
CM-v20-046	Raw pointwise detector ranks are not bottleneck-stable	proved	not_applicable
CM-v20-047	Finite-stencil nonzero soundness	proved	not_applicable
CM-v20-048	Birth detector zero-reflection	proved	not_applicable
CM-v20-049	Decoration-invariant germ detector zero-reflection	proved	not_applicable
CM-v20-050	Coverage-certified finite-stencil completeness	proved	not_applicable
CM-v20-051	Family-mesh detector completeness	proved	not_applicable
CM-v20-052	Exact augmented-mesh stencil count	proved	not_applicable
CM-v20-053	Family-relative MinCert upper bounds	proved	not_applicable
CM-v20-054	No uniform finite detector on an unbounded birth family	proved	not_applicable
CM-v20-055	Detector invariance under exact source-stage isomorphism	conditional	fixed_exact
CM-v20-056	Metric zero-verdict composition	conditional	fixed_metric_gap_certified
CM-v20-057	Canonical higher field-coefficient Ext edge	proved	not_applicable
CM-v20-058	Standard n -eligible higher Ext discharge	conditional	not_applicable
CM-v20-059	Natural zero-reflection upgrades discharge to equivalence	conditional	not_applicable
CM-v20-060	Synthetic duality does not create a natural bridge	proved	not_applicable
CM-v20-061	Representative invariance of higher Ext discharge	conditional	fixed_exact
CM-v20-062	Complete detector higher Ext discharge	conditional	not_applicable
CM-v20-063	Terminal and interior defect separation	proved	not_applicable
CM-v20-064	Complete transient-defect detector characterization	conditional	runtime_resolved
CM-v20-065	Joint terminal and long-transient classification	conditional	not_applicable
CM-v20-066	Conditional full-isomorphism upgrade	conditional	not_applicable
CM-v20-067	Terminal cleanliness is strictly weaker than full or metric cleanliness	proved	not_applicable
CM-v20-068	Detector-atlas zero requires global retained-bar coverage	conditional	not_applicable
CM-v20-069	Detector-aware Restart handoff is nonautomatic	proved	not_applicable
CM-v20-070	Eventually constant exact finite-to-infinite reduction	conditional	fixed_exact
CM-v20-071	Finite prefixes do not determine an infinite reduction	proved	not_applicable
CM-v20-072	Layered non-compensation for v20 certificates	proved	not_applicable
CM-v20-073	CM-v20 register closure and source-binding discipline	operational	not_applicable

16.2 Migration rule

The canonical UIDs CM-v20-001–CM-v20-040 conservatively succeed the same-numbered CM-v19 items at unchanged or weaker-safe mathematical strength. The former closure item CM-v19-041 is superseded by CM-v20-073. Item CM-v20-041 and items CM-v20-042–CM-v20-072 are new v20 records. No bare legacy identifier resolves without its theory-version namespace, and no v20 registration mutates a v19 source artifact or verdict.

17 Explicit exclusions

- (i) No global non-expansiveness, exactness, Serre localization, or strict global filtered functor for hard positive-threshold deletion.
- (ii) No metric transport through hard deletion without the two-sided threshold-gap hypotheses and the actual discrepancy bound.

- (iii) No repaired-zero transport without post-threshold zero separation; raw pointwise detector rank is not bottleneck-stable.
- (iv) No detector-zero conclusion from soundness alone, from a finite observation alone, or from an unverified birth, germ, stencil, mesh, family, endpoint, stage, or coverage record.
- (v) No family-relative MinCert claim reconstructed retrospectively from a single already-known target without the declared information class and noncircular evidence source.
- (vi) No uniform complete finite detector on a right-unbounded unbounded-birth family without additional stabilization, finite-birth, global-bound, cofinal, or finite-to-infinite structure.
- (vii) No higher PH-to-Ext equivalence without a complete standard n -eligible record and a separately proved natural zero-reflection theorem.
- (viii) No synthetic or toy shifted duality promoted to a natural realization theorem, representative-independent bridge, or external-domain theorem.
- (ix) No terminal Type IV zero interpreted as full defect vanishing, transient cleanliness, full isomorphism, or quantitative interleaving.
- (x) No defect detector reused across profile stages without reconstructing the actual kernel or cokernel profile and a new stage-specific certificate.
- (xi) No source/apex metric bounds automatically transported to kernel or cokernel defect profiles.
- (xii) No Restart reserve inequality interpreted as detector-zero transport, boundary coverage, higher coherence, or finite-to-infinite reduction.
- (xiii) No finite prefix interpreted as a complete countable atlas, tail theorem, eventual-constancy package, colimit certificate, or external Return Theorem.
- (xiv) No numerical slack manufactures a representative, morphism, eligibility record, naturality family, completeness proof, terminal-tameness proof, Claim UID, schema field, or artifact identity.
- (xv) No registration, successful build, package closure, replay, Search output, AI confidence, or human approval is mathematical proof by itself.
- (xvi) No normative Latin u for the Type IV cokernel diagnostic; v is canonical.

18 Primary literature and source regime

The imported single-morphism interleaving characterization is used only in the pointwise finite-dimensional one-parameter regime and is bound to U. Bauer and M. Lesnick, *Induced Matchings and the Algebraic Stability of Persistence Barcodes*, Journal of Computational Geometry 6(2) (2015), Corollary 6.6. All other v20 items are proved internally or extracted from the immutable Main and Foundation source claims listed in their metadata.

19 Provenance and status

This document was reconstructed from the submitted CM-v19 TeX source and audited against the canonical v20 Main, corrected Foundation A–G, and CR-v20 baseline artifacts. The inherited metric, crop, representative, Type IV, Restart, obligation, and single-morphism items retain their exact v19 guard-rails. The new v20 contribution is the reusable finite-certificate, higher-edge, transient-defect, detector-atlas, and finite-to-infinite reduction layer. CM-v20 remains internal Core theorem evidence only at the strength of each displayed statement and metadata record. CR-v20 promotion and TB-v20 consumption are subsequent governance actions.

Part IV. v21 provenance, impact boundary, and final status

Preservation. All seventy-three frozen CM-v20 theorem-bearing item identities remain defined exactly once, and the canonical registry contains one corresponding row for each item. The predecessor theorem corpus is not renumbered because all v21 additions use independent unnumbered headings and a separate v21 theorem counter.

Impact boundary. This revision changes no canonical statement in the v21 Main, Foundation A–G, Appendix DG, Appendix HT, or Appendix UB. It prepares the source-correct TB-v21 migration by replacing nine dangling historical references with explicit v21 compatibility edges. It does not retroactively repair or republish TB-v20.

Governance boundary. The local v21 items are ready for later global cross-audit. Their final Claim UIDs, global Migration/Ownership Matrix entries, release-gate effects, and package-acceptance status remain pending the final Claim Register, which is intentionally constructed last.

Document status: `mathematically_reaudited_and_build_validated`

Claim Register status: `pending_final_CR`

Release status: `pending_global_assembly_and_governance`

TB-v21 handoff: `consumer_contract_ready_with_dangling_reference_repair_map`